

Introduction

- The earth is the only known place in the universe where life exists. It is believed to be about **4.5 billion years** old.
- Life originated on it about **3.5 billion years ago** that is about one billion years after the earth came into existence.
- During this period there **evolved an environment** which supported life on it.
- It is this environmental complex which allowed and determined the **existence, evolution** and **survival** of innumerable forms of life on the earth.
- All these factors collectively and interactively form a **life-support-system** on the earth.
- Environment is thus a storehouse of various resources which are essential for life.

Introduction

- **Air, water, earth** and **biodiversity** are the major parts of our environment.
- Our life heavily depends on these interrelated components of environment. Any disturbance to any of these components will badly affect entire environmental complex and thus all humanity will have to suffer.
- Human interventions in the natural environment have already caused a great deal of detrimental changes in the environmental complex.
- **Global warming, ozone depletion, climatic change**, etc are some of the problems caused mainly due to man's undue interferences in the environment.
- It is now urgently important to design and adopt effective strategies and measures at local, regional and global levels to save natural environment for our own survival and sustainable development.

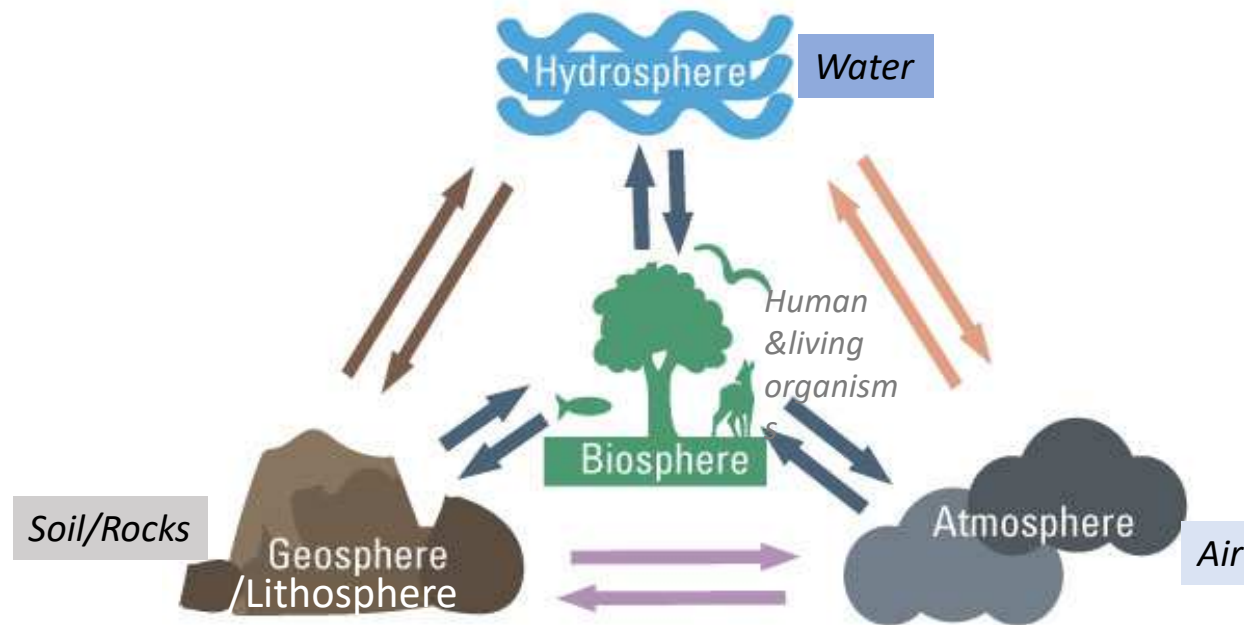
What is Environment?

- The word Environment is derived from the French word "**Environner**" which means '**surround**'.
- Environment is a word which describes **an aggregate** of all the **external forces** influences and conditions (living or non-living), which affects the human life and its existence on this planet.
- It covers all the outside factors that have acted on the individual since he/she/they/it began life.
- Thus, environment is the aggregate of physical, chemical, biological, economical and social components on Earth which are capable of causing **direct or indirect effects** on **living and non-living things** for a **shorter or longer duration** and their interactions and activities.
- **Multi-disciplinary** in nature.
- It is **not static** and tends to change continuously.

Definition:

Environment is everything surrounding us and is defined as "The social, cultural, physical, geographical, chemical, biological and ecological conditions that **surround, affect and influence** the survival, growth and development of all living organisms".

Concept of Environment



- Air, Water, Land, living organisms and materials surrounding us and their interactions together constitutes environment.

Goals/Objectives



- Creating the awareness about environmental problems in the people mind.
- Giving basic knowledge about the environment and its related problems.
- Providing information regarding modern environmental concepts, conserving biodiversity, how to lead more sustainable lifestyles and how to use resources more equitably.
- Developing an attitude of concern about the environment amongst the people.

Goals/Objectives



- Acquiring skills to help the concerned individuals in identifying and solving environmental problems
- To change the way in which people view their own environment by a practical approach based on observation and self learning.
- To create a concern for environment that will trigger/motivate pro-environmental action, including activities they can do in their daily life to protect it and improve it.

Background of the Environmental Studies subject in India

- In modern India, our late prime minister **Mrs. Indira Gandhi** has played a major role in introducing the concept of environmental protection in the constitution of India as a fundamental duty under **Article 51A(g)**- It is clearly mentioned that it is the duty of every citizen to protect and conserve our environment and natural resources.
- Justice **Kuldeep Singh**, known popularly as the '**Green Judge**' and **Shri M.C. Mehta** the '**Green Advocate**' of India have greatly contributed for the environmental education.
- **In 1991** the Supreme Court of India issued directives to make all the curricula in which **Environmental Studies should be included** as per the directives of the Supreme Court.

Types of Environment



Natural Environment



**Artificial or
Manmade Environment**
(Economic, Social and Cultural)



Types of Environment

- However, on the basis of proximity or nearness and the nature of the influences exerted on the organisms by the environment it can be of following types:
 - a. **Micro-environment:** It refers to the immediate local surroundings of an organism.
 - b. **Macro-environment:** It refers to all the physical and biotic conditions that surround the organism externally. Microenvironment is merely a part of it.
- Similarly on the basis of the composition or nature of the environmental components following types of environment can be enumerated.
 - a. **Terrestrial environment** where a major components of environment is land
 - b. **Aquatic environment** where water forms the major portion of an individual's environment
 - c. **Aerial environment** where air is the dominant environmental factor

The classification of various components of environment have been done in the following way:

❖ **Physical Environment or Physical component of environment**

refers to all abiotic factors or conditions like soil, minerals, temperature, light, rainfall, etc. It comprises of atmosphere, lithosphere and hydrosphere.

❖ **Biological environment or Biological component of environment**

includes all biotic factors or living forms like plants, animals and micro-organisms.

❖ **Social environment or Social component of environment**

includes an individual's social, economic and political condition wherein he lives. The moral, cultural and emotional forces influence the life and nature of individual behavior.

Components of Environment:

Physical:

Atmosphere
Hydrosphere
Lithosphere } **Abiotic**

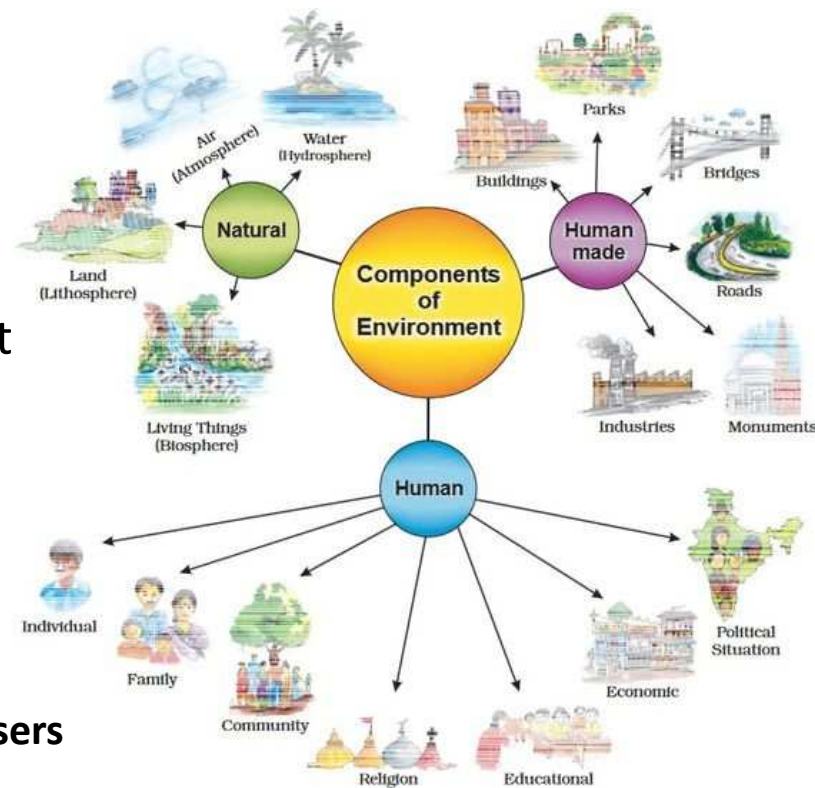
Biological:

Biosphere: All living things present in Atmosphere, Hydrosphere and Lithosphere

Biotic

Producers Consumers Decomposers

Social: Customs, culture, habits, income, occupation, religion etc.



a. The Atmosphere (atmos=air; sphere-orbit, area)

- It refers to **the mantle of gases** which surround our planet.
- It is a complex mixture of a number of **gases, water vapours** and a variety of **fine particulate** material.
- Most of these gases are compressed in the lowermost layer.
- It also acts as a blanket that makes life possible on the earth by regulating temperature, absorbing toxic gaseous wastes, filtering harmful solar rays and facilitating water cycle.
- The thin layer of gases that envelops the Earth is held in place by the **planet's gravity**.
- Dry air consists of approximately **78% Nitrogen, 21% Oxygen, 1% Argon and other inert gases**.

a. The Atmosphere (atmos=air; sphere-orbit, area)

- **Other gases** present in earth's atmosphere are referred to as **trace gases**.
- Air also contains a variable amount of water vapor and suspensions of water droplets and ice crystals seen as clouds.
- Many natural substances such as dust, pollen, spores, volcanic ash, etc may also be present in tiny amounts in air.
- The ozone layer of the Earth's atmosphere plays an important role in depleting the amount of ultraviolet (UV) radiation that reaches the surface.

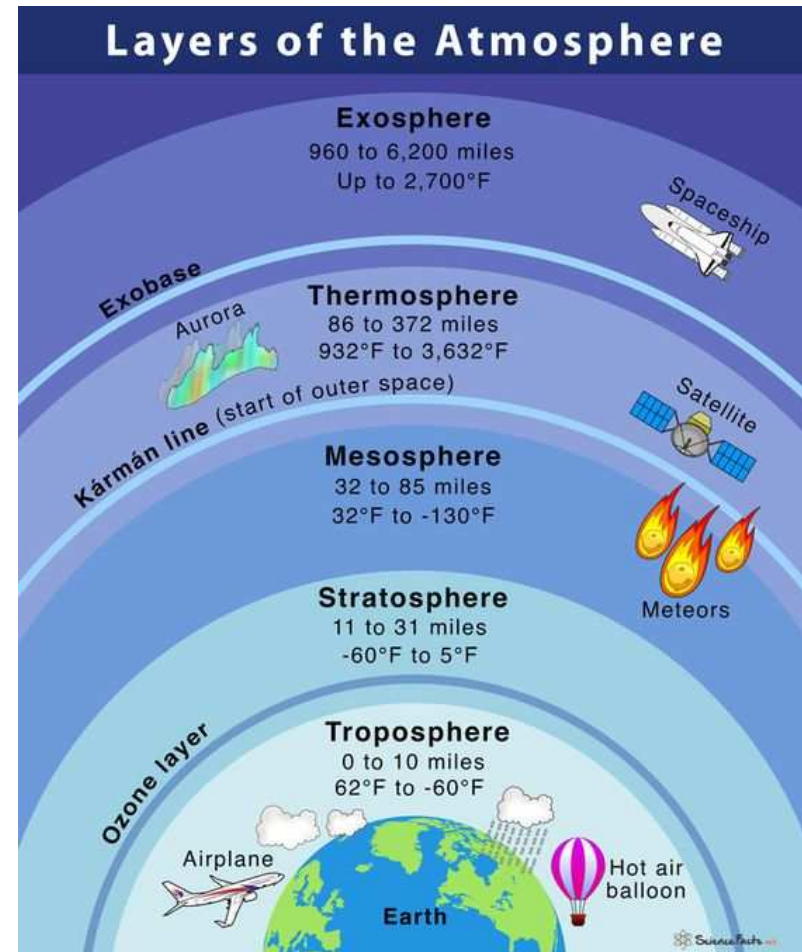
Gas	Per cent by Volume
Nitrogen	- 78.08
Oxygen	20.95
Argon	0.93
Carbon Dioxide	0.03
Helium	0.00052
Neon	0.0018
Krypton	0.00010
Methane	0.00015
Hydrogen	0.00005
Nitrous Oxide	0.00005
Xenon	0.000009
Ozone	0.000007

Structure or layers of the Atmosphere:

- **The Atmosphere is divided into different layers which show different patterns of temperature.**
 1. Troposphere
 2. Stratosphere
 3. Mesosphere
 4. Thermosphere/Ionosphere
 5. Exosphere
- **Why do we need atmosphere?**
 - ❖ All climatic changes occur in it.
 - ❖ Water and other geochemical cycles need this component to complete.
 - ❖ Oxygen and other important gases which are required for life are present in it.
 - ❖ It acts as a sink for pollutants and other materials.
 - ❖ It protects life on the earth from harmful radiations.

Atmosphere

- The term 'atmosphere' comes from Greek word '*atmos*' meaning **steam or vapor** and **sphere or globe**.
- **The layer of gases** that surrounds the earth is known as Atmosphere.
- It allows us to breath and protects earth from harmful radiations.
- The atmosphere is composed of **4 layers based on temperature**.
- They are 1. Troposphere 2. Stratosphere 3. Mesosphere 4. Thermosphere and outer one called Exosphere



Atmosphere Layers

Region	Altitude range (km)	Temperature range (C)	Important chemical species
Troposphere	0-20	15 to -56	Nitrogen, oxygen, water vapour, carbon dioxide
Stratosphere	20-50	-56 to -2	Ozone
Mesosphere	50-85	-2 to -92	Oxygen, nitric oxide
Thermosphere	85-500	-92 to 1200	Oxygen, nitric oxide
Exosphere	500-10,000+	0 to 1700	H, He, CO ₂ , O ₂

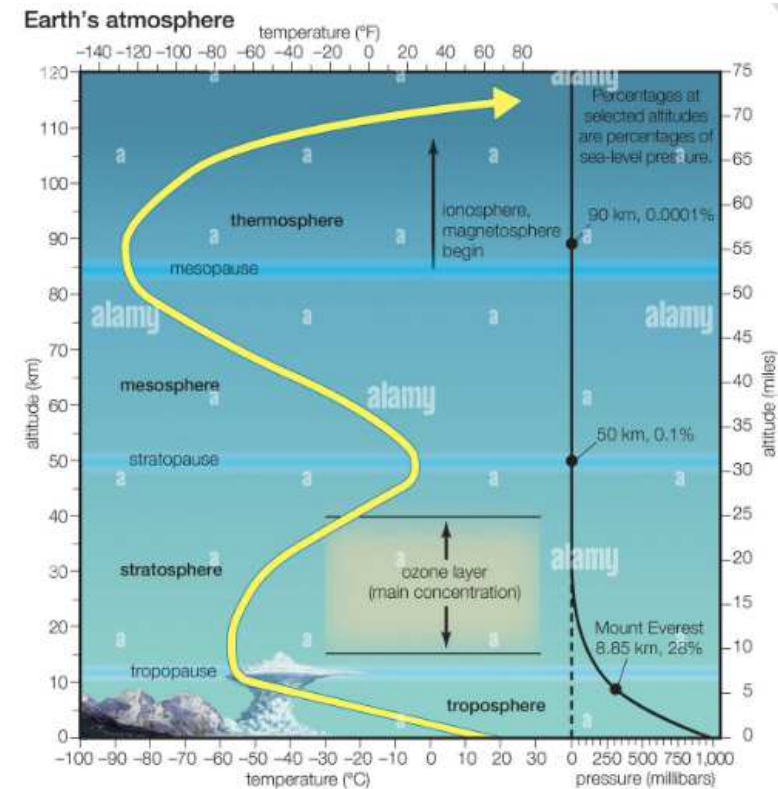
Vertical Structure of Atmosphere

- Atmosphere is gaseous envelope that surrounds the celestial body. Therefore, the earth like other planets in solar system has an atmosphere which is retained by gravitational attraction and largely rotates with it.

Weather: Day to day changes in the atmosphere

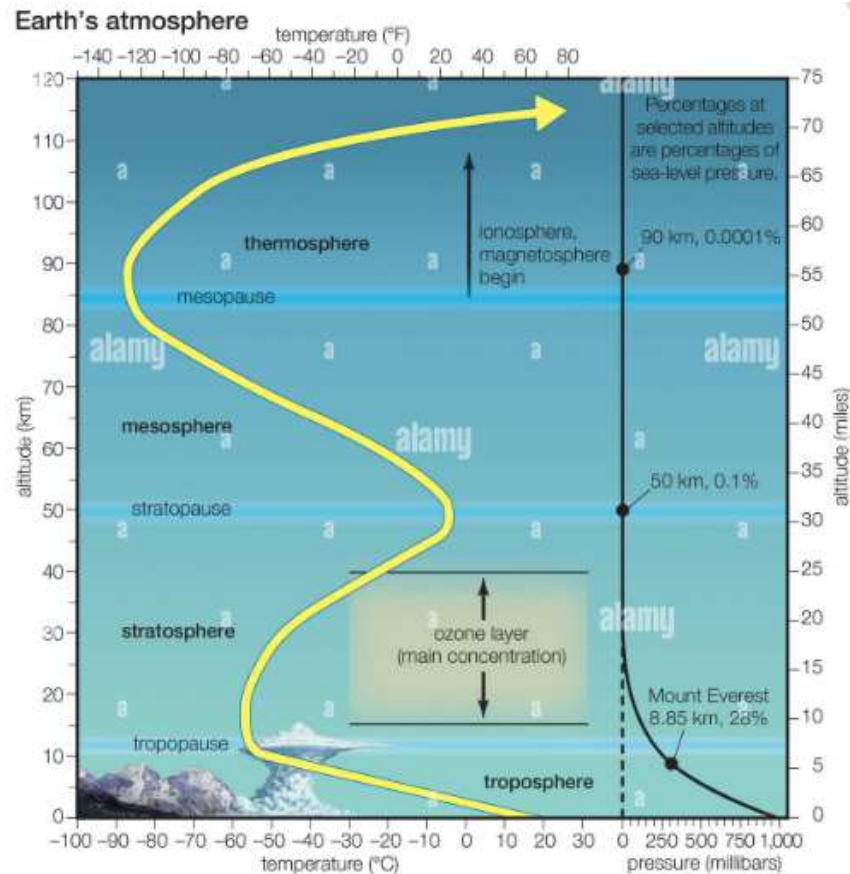
Climate: Average weather conditions over a longer period - 30 years

- Earth Atmosphere has divided into vertical layers according to distinctive physical properties. These are **thermal characteristics**, **chemical composition**, **electrical attributes**, or **density**.



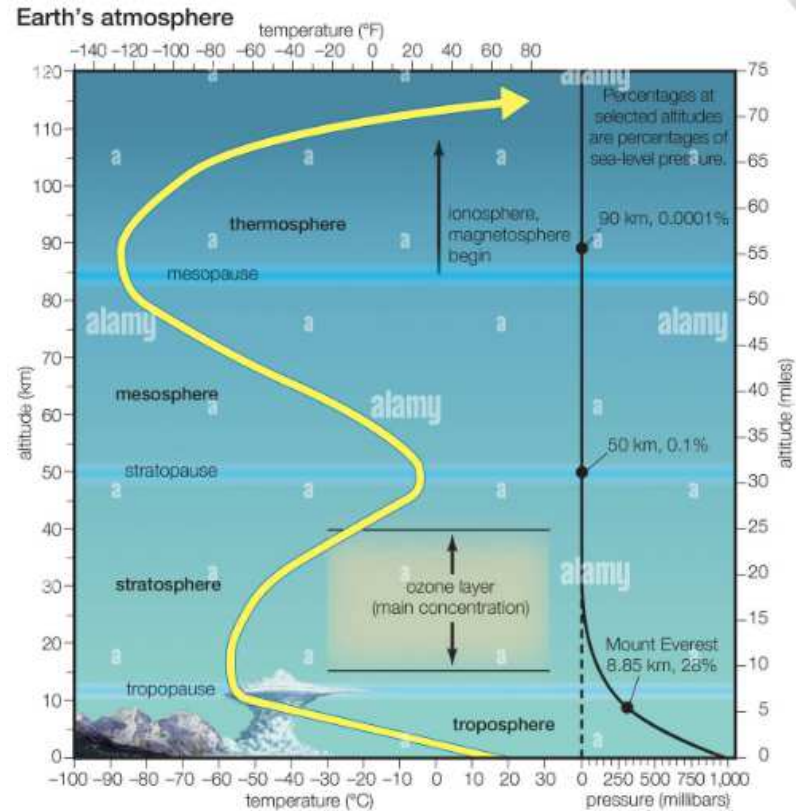
Troposphere:

- First layer of atmosphere. It is the lower most region of atmosphere which is in contact with earth's surface is called **troposphere**.
- It is up to maximum 18km from the ground surface near equator and 7 to 10 km near poles pressure.
- All weather changes take place in this layer - clouds, rain, snow.
- Temperature decreases with altitude.

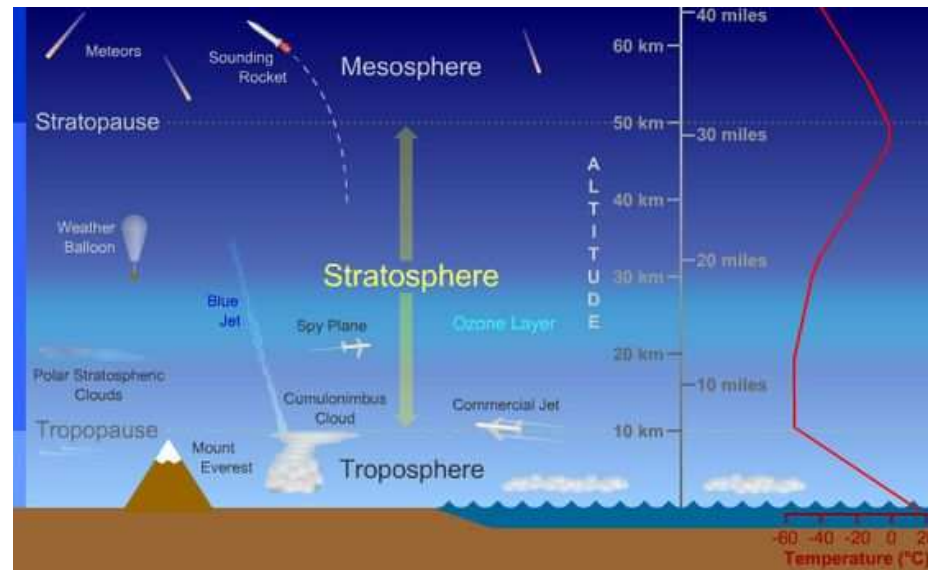
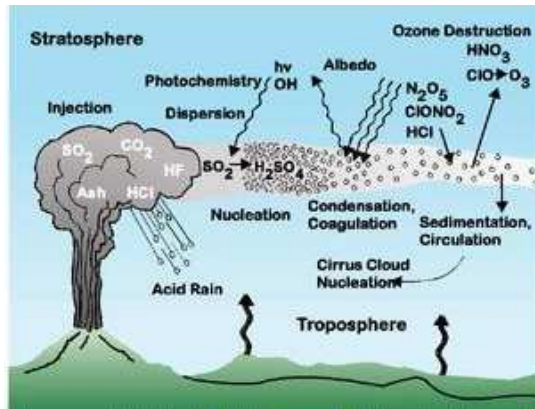


Troposphere:

- Temperature varies from 15°C to -56°C . The **temperature** in this layer drops with height.
- The uppermost boundary of this layer is called **Tropopause** demarcates between the troposphere and stratosphere.
- Temperature gets colder as distance from the earth increases by about $6.5^{\circ}\text{C}/\text{km}$, called as **lapse rate**.
- It contains approximately about 75% of atmosphere.



Stratosphere:

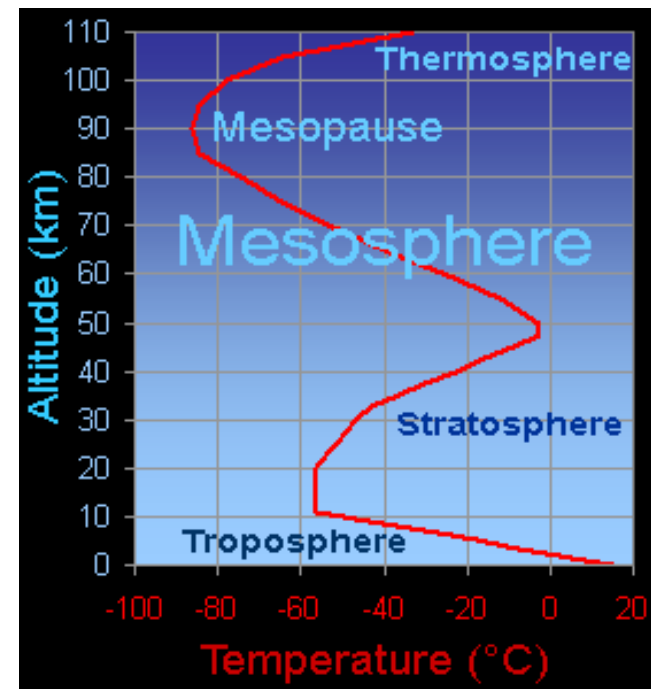
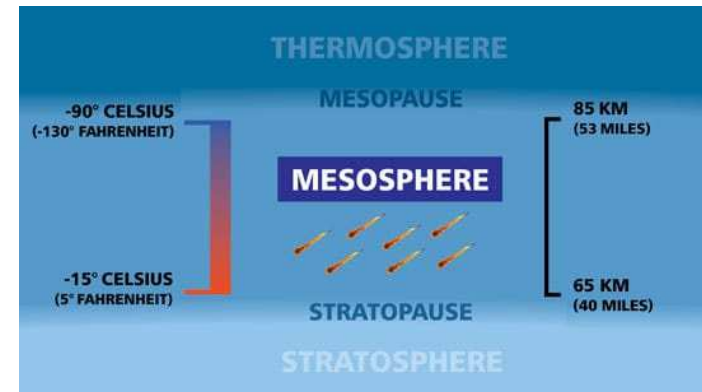


- This is the layer next to troposphere. **Second layer of atmosphere.**
- It extends upto 50 km from ground surface and is about ~ 30 km in thickness.
- It is very important zone of atmosphere as it contains the **vital ozone layer**.
- Temperature in this zone is first constant till 20km then rises with height to 0°C at its upper extremity due to absorption of UV radiation from the sun by ozone gas.
- The ozone layer is mainly located in the lower portion of the stratosphere from about 15-35 km, though the thickness varies seasonally and geographically.

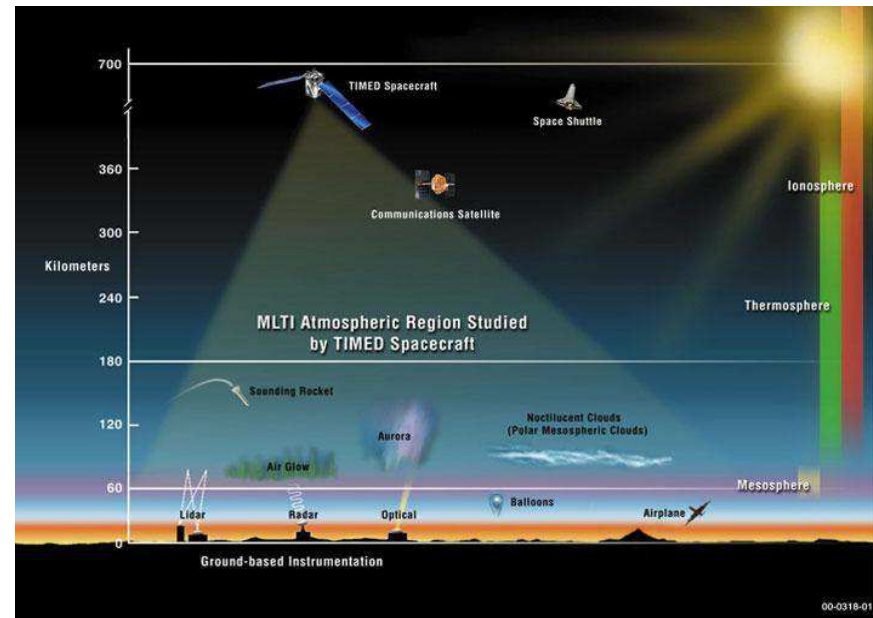
- About 90% of the ozone in our atmosphere is contained in the stratosphere.
- It protects us from skin cancer.
- Stratosphere also contain sulphate particle aerosols which are important for rain formation.
- The boundary between the stratosphere and mesosphere is called as the **stratopause**.

Mesosphere

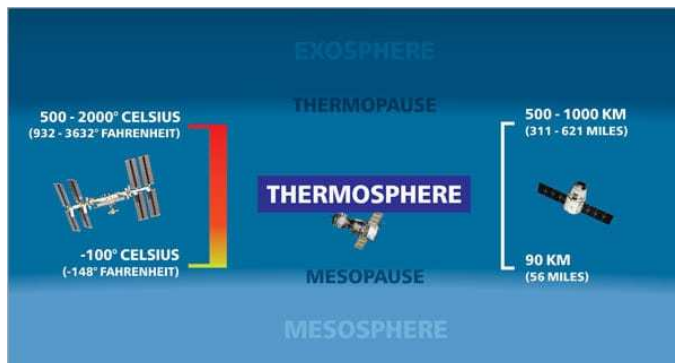
- **Meso** means **middle**, also called middle atmosphere, the **3rd layer**.
- It begins 50 km above earth and ends about 85km.
- Above the stratosphere temperature begins to drop from the beginning of the mesosphere as there is no water vapor, dust, clouds and O3.
- Atmospheric pressure is very low i.e. less than 1% of the pressure at sea level.
- Air is very cold and temperature varies **-2°C to -92°C at Mesopause** (coldest part of the atmosphere).
- Protects the earth surface from being hit by most meteoroids.



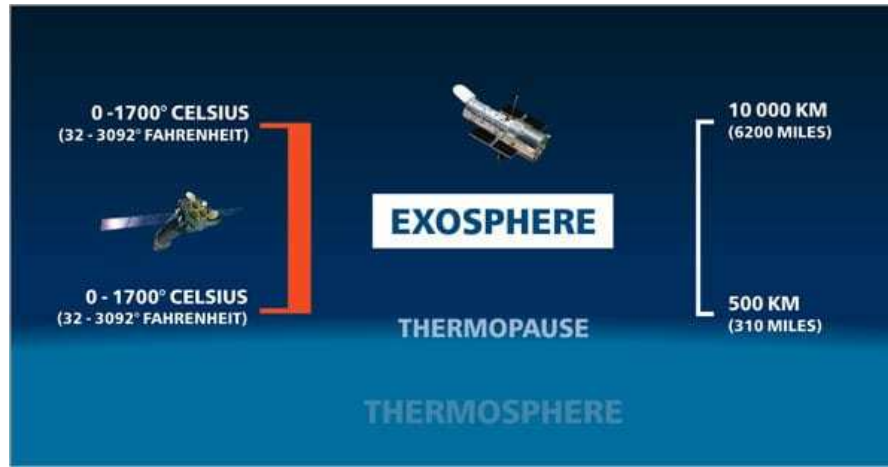
Thermosphere



- Also known as **ionosphere** (hot layer).
- This occurs from 85km to 500km from the ground
- This contains thin layer of gases which absorbs much of the sun light UV and X-rays causing gaseous molecules to split into atoms and ions, is significantly warmer (500°C to 2000°C) than other layers.
- **Auroras** (northern lights and southern lights) occur and the space shuttle orbits in this layer.



Exosphere



- The region above 500 km.
- **Hydrogen is present throughout** the exosphere, with some **Helium, CO₂ and O₂ near its base.**
- It is hard to define the boundary between the exosphere and outer space, the exosphere may be considered a part of the interplanetary space or outer space.
- Very little is known about it due to lack of research.

b. The Hydrosphere (hydro-water; sphere-domain, area)

- Hydrosphere refers to the total mass of water on the earth.
- The hydrosphere includes water that is on the surface of the planet, underground, and in the air.
- It, thus, can be liquid, vapour or ice. Liquid water on the earth exists on the surface in the form of oceans, lakes and rivers. It also exists below ground-as groundwater, in wells and aquifers. Water in vapour form is most visible as clouds and fog.
- The frozen part of Earth's hydrosphere is ice that is found in glaciers, ice caps and icebergs.
- This frozen part of the hydrosphere is also known as the cryosphere.
- Water is an absolute essentiality of life on our planet.
- It is a natural resource of fundamental importance and its properties are very primarily useful to sustain life and life support system on the earth.

b. The Hydrosphere (hydro-water; sphere-domain, area)

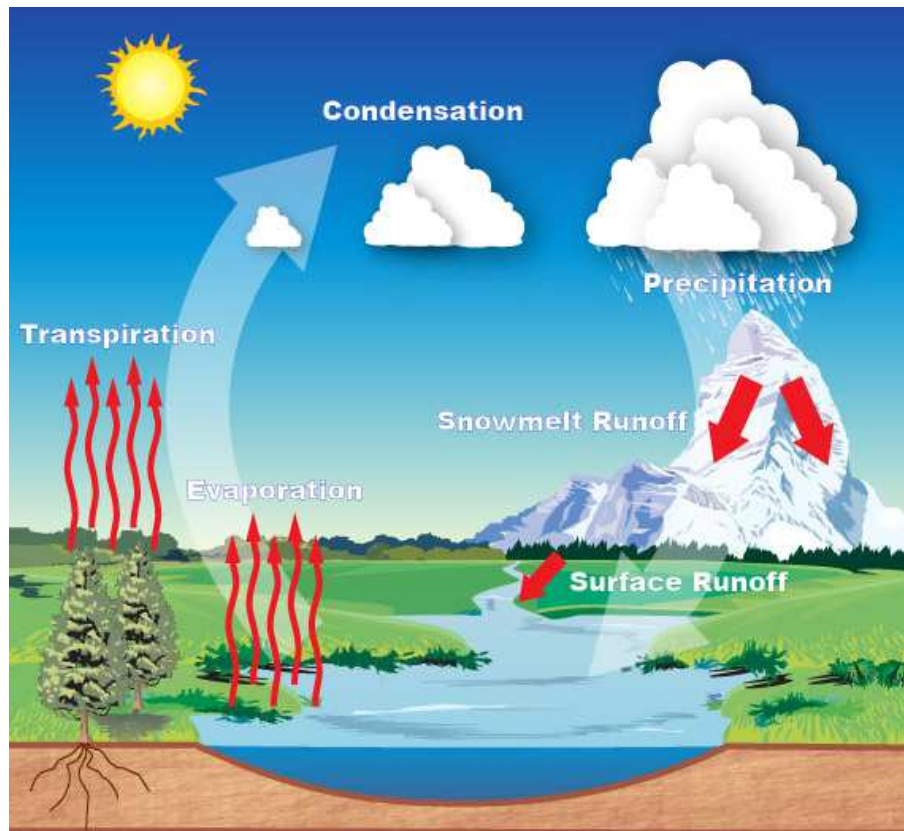
- Water is the medium in which all the **biochemical process** within a living organism occur.
- Outwards, it is the availability of water which determines the nature, composition and abundance of the terrestrial life.
- Climatic changes on the planet are primarily linked with water. Of the total estimated water about **95% is locked in the lithosphere** and only about **5% is in free circulation**.
- Out of this free water 97% is in oceans.
- Major portion of the earth's crust (about 71%) is occupied by water.
- Oceanic water is saline. Its average salinity is around 35 parts per thousand (or 3.5%).
- Both **atmosphere and hydrosphere** are always in motion.
- Water moves through the **hydrosphere** in a cycle known as **water or hydrologic cycle**. Water collects in clouds, and then falls to Earth in the form of rain or snow.
- This water collects in rivers, lakes and oceans. Then it evaporates into the atmosphere to start the cycle all over again.

Distribution of free water on our planet

- Total water quantity of hydrosphere is approx. 1360 million km³.
- The water content is classified into Salty water and Fresh water
- 97.2% is salty water (present in oceans as marine ecosystem) and only 2.8% is available as fresh water.
- The 2.8% available fresh water exists as glaciers (2.15%), underground water (0.62%) and remaining as surface water (0.02%) in rivers, lakes and ponds.
- Human activities such as deforestation create serious changes.

The Component	Water as % of total water
1. Oceans	97.2
2. Ice sheets and glaciers	2.15
3. Ground Water	0.62
4. Rivers, lakes and inland seas	0.020
5. Soil moisture	0.005
6. Atmosphere	0.001

Hydrologic cycle



● Special properties of water

1. cohesion & adhesion
 - surface tension, capillary action
2. good solvent
 - many molecules dissolve in H₂O
 - hydrophilic vs. hydrophobic
3. lower density as a solid
 - ice floats!
4. high specific heat
 - water stores heat
5. high heat of vaporization
 - heats & cools slowly

Boiling point - 212°F (100°C)

Melting point - 32°F (0°C)

Chemical formula - H₂O

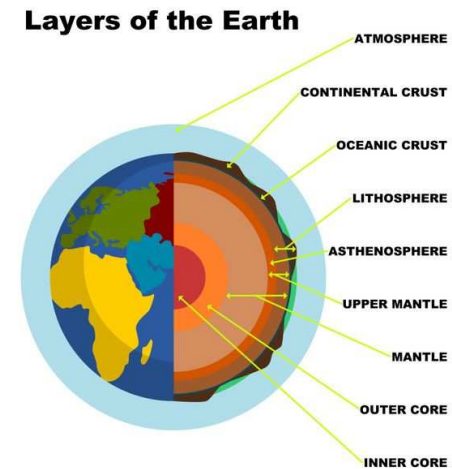
Average Molar mass - 18.01528 g/mol

IUPAC ID - Water, Oxidane

Density - 997 kg/m³

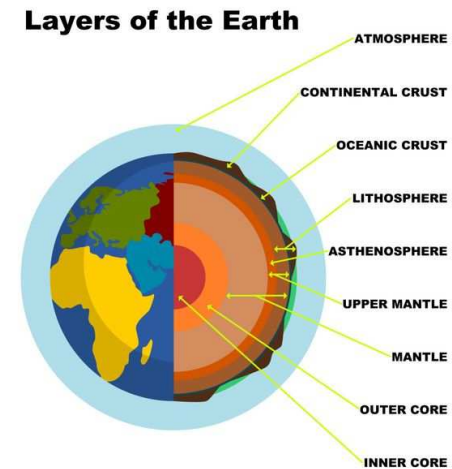
c. The Lithosphere (lithos = rock; sphere-area, domain)

- The earth is divided into various regions or layers from its innermost center to its outer surface.
- The innermost layer is called as Core, whereas middle region is named as **Mantle** and the outermost part of the earth is known as **Crust**.
- The **lithosphere** refers to the solid, outer part of the earth. It includes the brittle upper portion of the mantle and the crust.
- It is bounded by the atmosphere above and the **asthenosphere** (a part of the upper mantle) below.

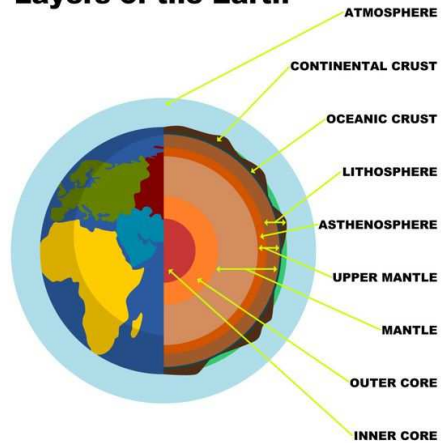


c. The Lithosphere (lithos = rock; sphere-area, domain)

- The lithosphere is the most rigid of Earth's layers and extends from the surface of the earth to a depth of about 70-100 km. There are two types of lithosphere viz. ***oceanic lithosphere*** (which is under oceans) and ***continental lithosphere*** (which is open or terrestrial landmass).
- Entire lithosphere is composed of several crustal plates known as **tectonic plates** which keep floating on the molten material below them.
- Movements of these tectonic plates bring about various changes on earth's surface such as earthquakes, volcanic activity and continental drifts.

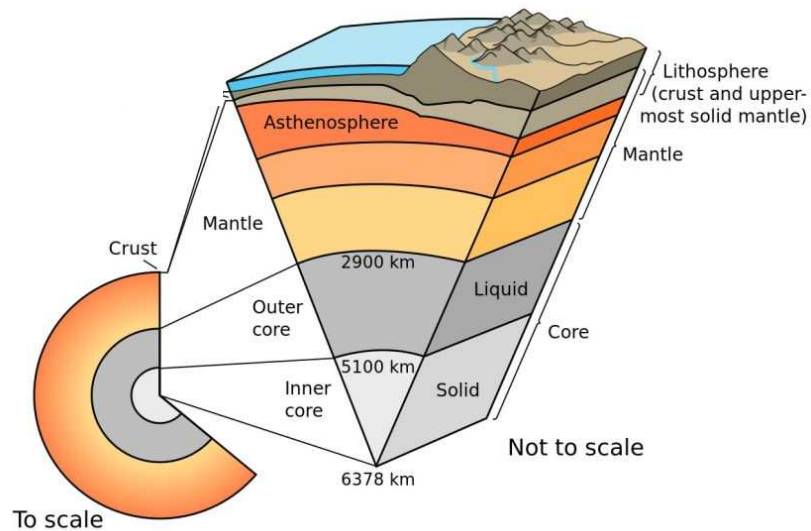


Layers of the Earth

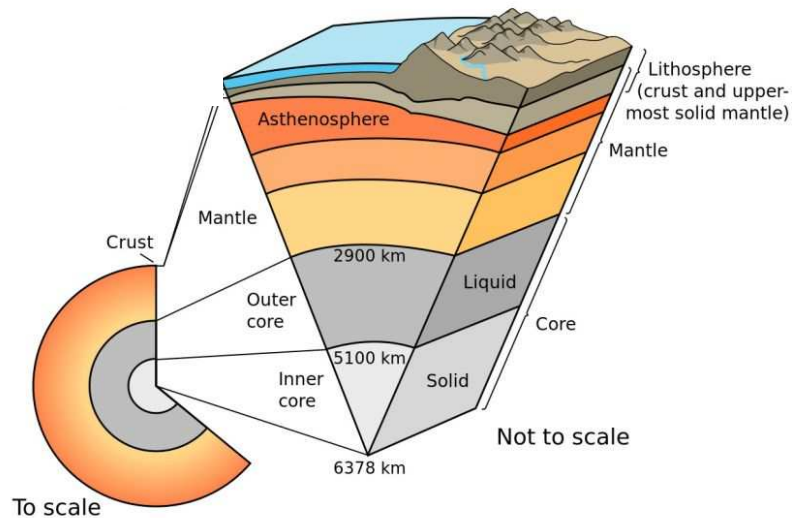
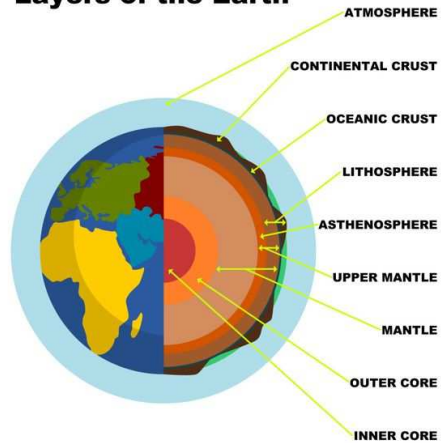


Lithosphere

- Upper 100 km thick layer of earth consisting of Crust and some of upper Mantle is called **lithosphere** (from surface to a depth of 100km).



Layers of the Earth

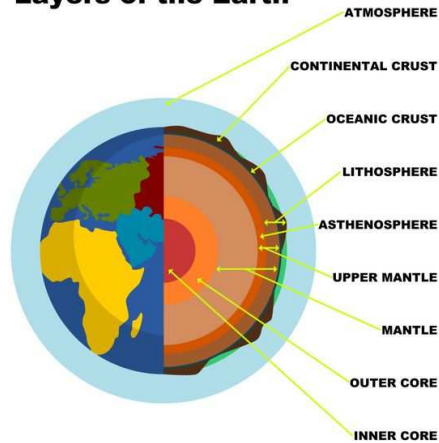


Lithosphere

Earth Structure:

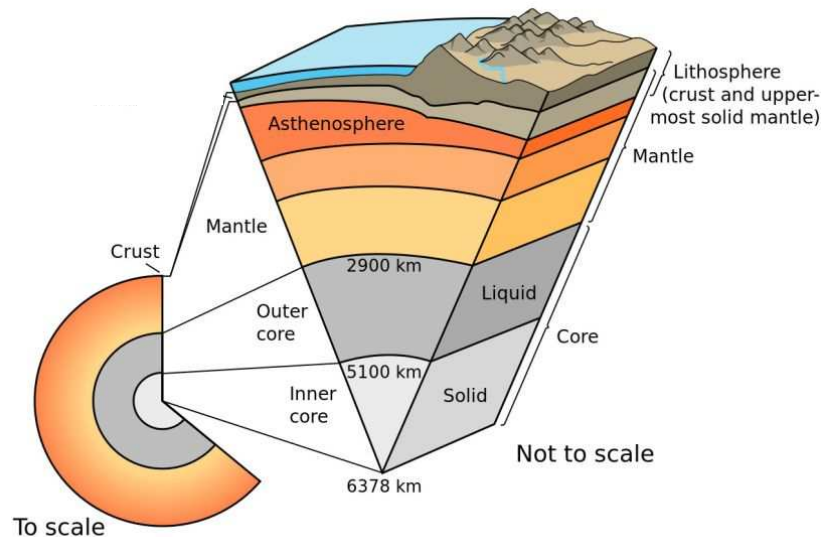
- **Crust** - uppermost surface of earth. The depth of crust is about 5 to 40 km, with average of 17 km. Crust is made of silicon and oxygen than Fe and Nickel. The crust is heavier and denser in the oceans (oceanic crust), lighter and thicker at continents (continental crust). (47%-O₂; 27%-Si; 8%-Al; 5%-Fe; 4%-Ca; 2%-Mg, K, Na).
- Rocks when broken down, forms soil on which man is dependent for his agriculture.
- The minerals present in earth crust are the raw materials used in various industries.

Layers of the Earth

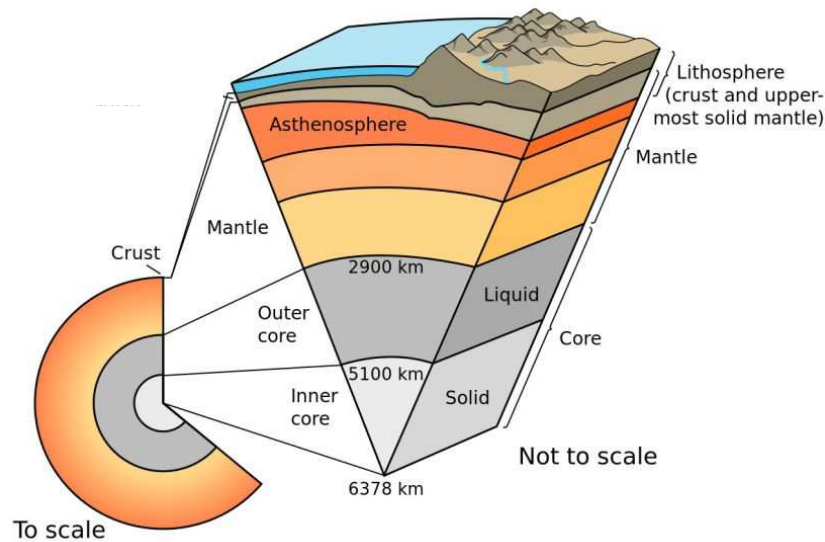
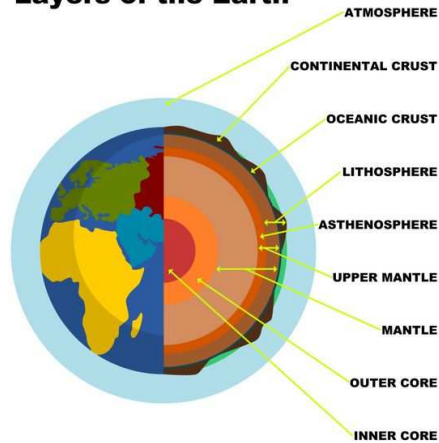


Lithosphere

- **Mantle** - below crust and depth of 2900 km. Mantle is made of rocks which are less dense than Fe and Ni.
- The upper mantle is made up of the **asthenosphere** (80-200 km depth from the surface) and part of the lithosphere above that (the hardened rocks of the upper mantle and the crust which forms all the tectonic plates of the earth).
- The Lower mantle is heated by Core and is semisolid and able to move and slide around.



Layers of the Earth

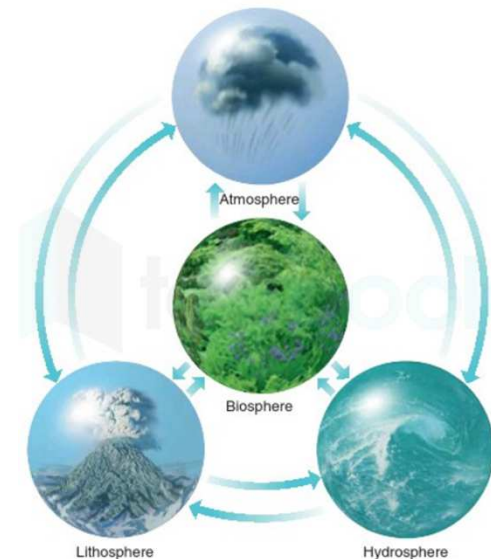


Lithosphere

- **Core** - inner to mantle is core divided into outer core of 2250 km and inner core of 1200 km.
- The **solid inner core** is made of Iron and Nickel. Although the inner core is intensely hot (5430°C), it is also under extreme pressure and is therefore unable to melt.
- The **outer core** has less weight on top of it and therefore is liquid or molten, melted iron and nickel. Because of movements of the liquid metal of the outer core, the earth generates its own magnetic field, which changes shape and position over time. Total radius of earth is 6378 km.

B. Biological Environment or Biosphere (bio-life; sphere-area, domain)

- It refers to the **zone of life on Earth**.
- It, in fact, relates to the living organisms which are part of an individual's environment or surroundings.
- Every living organism is surrounded by other living organisms which are part of its environment.
- Just like physical factors of environment, the other living organisms present in surroundings also affect the life and survival of an individual.
- Man is surrounded by billions of other living organisms which play very important role in his life. Plants, animals and microorganisms present on the earth affect man's life in innumerable ways.
- **Biosphere** is postulated to have evolved, at least some 3.5 billion years ago.
- The term "**biosphere**" was coined by geologist **Eduard Suess** in 1875, which he defined as '**the place on Earth's surface where life dwells**'.

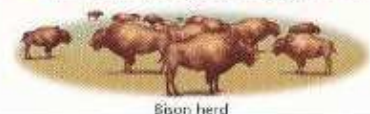
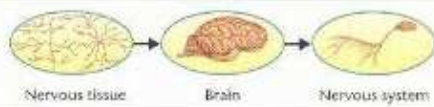


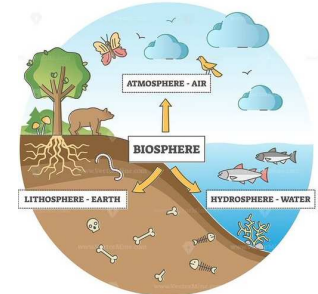
All the components of environment are, interlinked, interactive, interdependent and overlapping

Biosphere

- All the living creatures present on the earth are collectively known as ***Biosphere***.
- It is also known as **the sum total of life on the earth**.
- There are millions of species of living organisms belonging to a large number of families and taxonomic groups. Some estimates say there are 15,95,225 species present in the biosphere.
- **Wilson** in 1992 put the total number of living species in a range of 10 million to 50 million.
- Biological environment can further be classified in various ways. Depending upon the nature of organisms, they can be classified as **plants, animals, bacteria, fungi, viruses, etc.**
- On the basis of their role and status in the ecosystem, they can be categorized as **producers** (green plants), **consumers** (animals) **and decomposers**.

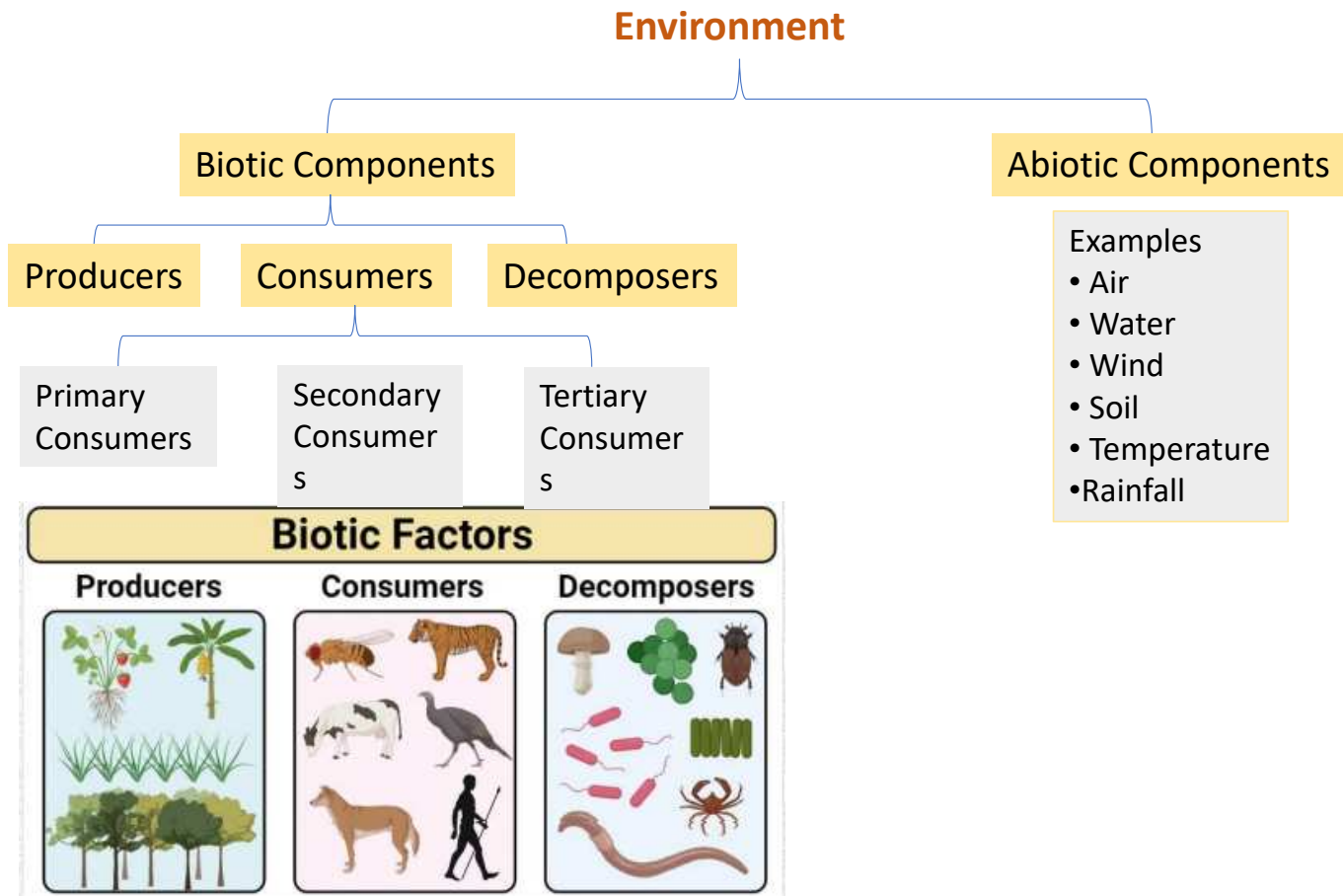
Biosphere (Biological Component of Environment)

Biosphere	The part of Earth that contains all ecosystems	
Ecosystem	Community and its nonliving surroundings	
Community	Populations that live together in a defined area	
Population	Group of organisms of one type that live in the same area	
Organism	Individual living thing	
Groups of Cells	Tissues, organs, and organ systems	
Cells	Smallest functional unit of life	
Molecules	Groups of atoms; smallest unit of most chemical compounds	

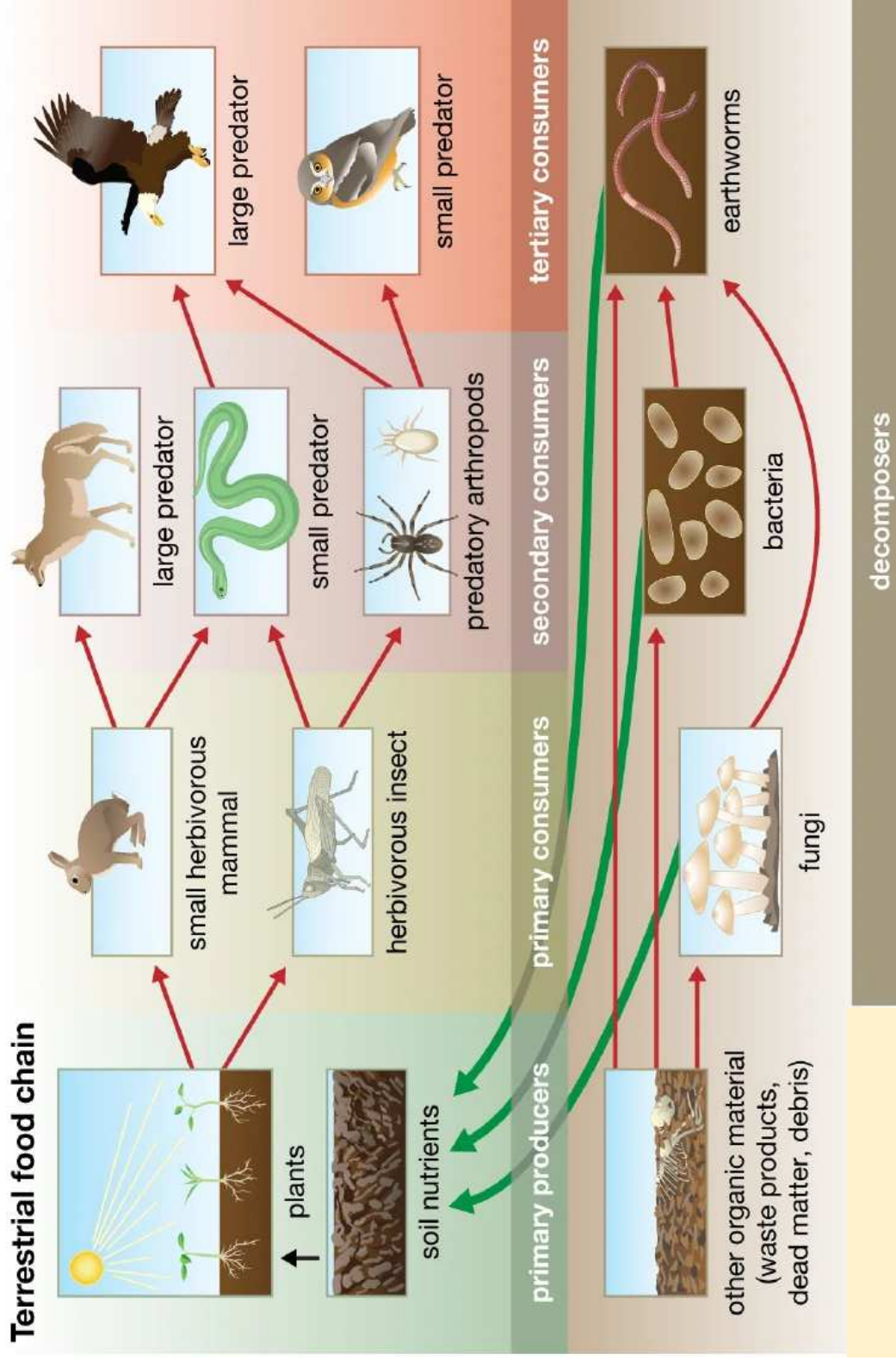


- We all know that life exists in **atmosphere, hydrosphere and lithosphere**. There is interactions of life with all the 3 physical components of Environment.
- Life exists in **atmosphere** up to an altitude of **8 km** from the ground surface.
- **Upto a depth of 8 km in ocean/water** and in **lithosphere life exists up to where the root penetrates the soil**.
- The concept of Biosphere is like this - group of Molecules become Cells, group of Cells become Tissues and organs and Organism-group of similar organisms called Species-group of Species become Community-Community and its interactions with non living surroundings like abiotic factors water, soil and air makes the Ecosystem.
- Different types of ecosystems on the earth makes the **Biosphere**.

Biotic components



Terrestrial food chain



C. Social Environment

- Social environment refers to the immediate social setting in which people live.
- It includes the cultural, educational, political and other social aspects of a society where an individual lives.
- It also includes the aspects of social interaction including its products such as beliefs, attitudes, traditions, cultural values, etc.
- Various aspects of a man's social living affect his individuality, health, perceptions and overall welfare.
- Education level, socioeconomic status, awareness level, social interactions and moral values prevalent in a society collectively form the social environment of an individual.

Factors of social environment

- Culture and history
- Standards of living
- Education and awareness
- Socioeconomic conditions
- Social stability
- Moral values in the society
- Political structure
- Economic system
- Technological application
- Availability of basic amenities

All these social factors have great bearing on general personality, individuality, health and progress of an individual in the society.

Sustainable Development



Sustainable development is defined as, '*development that meets the needs of the present without compromising the ability of future generations to meet their own needs*'.

Sustainable Development

- Generally **development means economic progress**. But this economic progress is often achieved at the cost of long term losses. These losses include environmental implications and depletion of resources which will not be available for future generations.
- A clear discussion on sustainable development emerged on an international level in 1992, in the UN Conference on Environment and Development (UNCED), popularly known as **The Earth Summit**, held at Rio de Janeiro, Brazil.

Sustainable Development

- The Rio Declaration aims at “a new and equitable global partnership through the creation of new levels of cooperation among states.”
- Out of its five significant agreements **Agenda21** proposes a **global programme of action on sustainable development** in social, economic and political context for the 21st Century.
- Sustainable development is described in terms of **three dimensions**, which include “**economic, environmental and social**” aspects. Some experts adds another fourth dimension to it that is ‘**culture or governance**’.

Concept of Sustainability

- **Definition:**
 - ✓ “**Sustainability** means meeting our own needs without compromising the ability of future generations to meet their own needs”
 - ✓ The definition was given by **G.H. Brundtland** in 1987, who was Norway prime minister and president of WHO.



- *The concept of sustainability is composed of three pillars: **economic, environmental, and social**, also known informally as **profits, planet, and people**.*

Sustainable Development



To achieve SD there are 3 principles:

1. Improving the quality of human life
2. Economic growth for all
3. Environmental development

Definition:

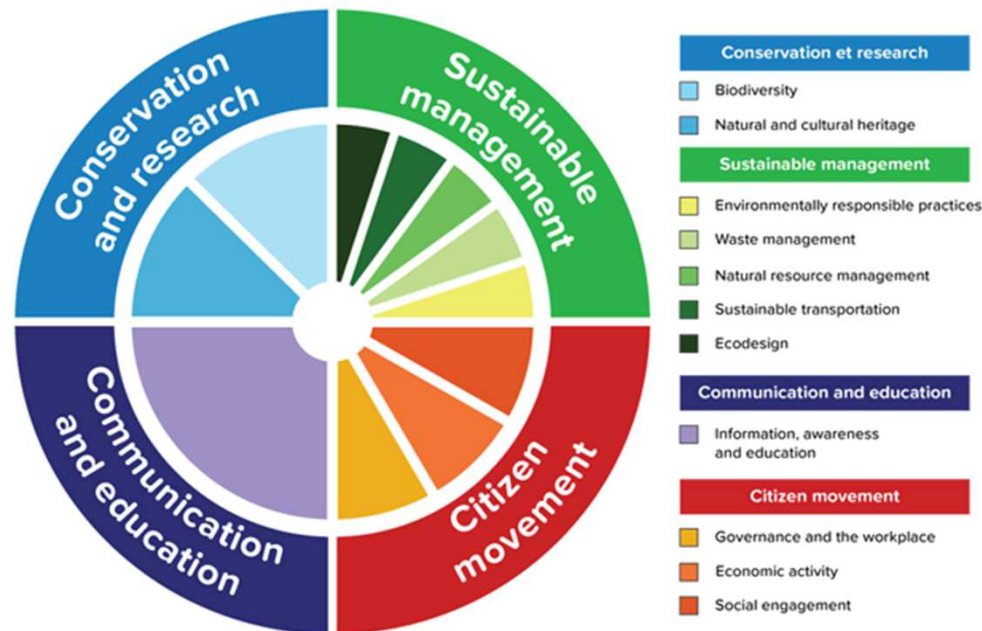
✓ Development which enables individuals and communities in underdeveloped regions of the world to raise living standards through profitable products, consistent with minimizing adverse environmental effects.

Objectives of Sustainable Development

- **Objectives:**

- 1) To protect our Biodiversity
- 2) To increase forest cover
- 3) To prevent pollution
- 4) To reduce, reuse and recycle waste
- 5) Design eco-friendly technology
- 6) To control population growth

Sustainable Development Policy



The goal of this approach is to minimize our impact on the environment while improving harmony and quality of life within the organization and promoting equality between individuals, populations and generations.

Sustainable Development

- From **environmental point** of view development is said to be sustainable if it does not harm environment and natural ecosystems.
- **There are some indicators of sustainable development such as:**
 - ❖ Consideration of possible environmental impacts due to developmental projects
 - ❖ No or minimum damage to environment and natural ecosystems
 - ❖ Use of appropriate and eco-friendly technology
 - ❖ Adoption of **Reuse, Recycle and Reduce (3Rs)** approach
 - ❖ Minimum or no waste generation
 - ❖ Raising environmental awareness and education

Sustainable solutions exist:

- **We must develop solutions that protect both our quality of life and the environment**
- Organic agriculture
- Reduction of pollution
- Protecting biodiversity
- Waste disposal
- Recycling
- Alternative fuels



The Sustainable Development Goals

- **The Sustainable Development Goals** or Global Goals are a collection of 17 interlinked global goals designed to be a "blueprint to achieve a better and more sustainable future for all"
- The Sustainable Development Goals (SDGs), also known as the Global Goals, were adopted by the United Nations in 2015 as a universal call to action to end poverty, protect the planet, and ensure that by 2030 all people enjoy peace and prosperity

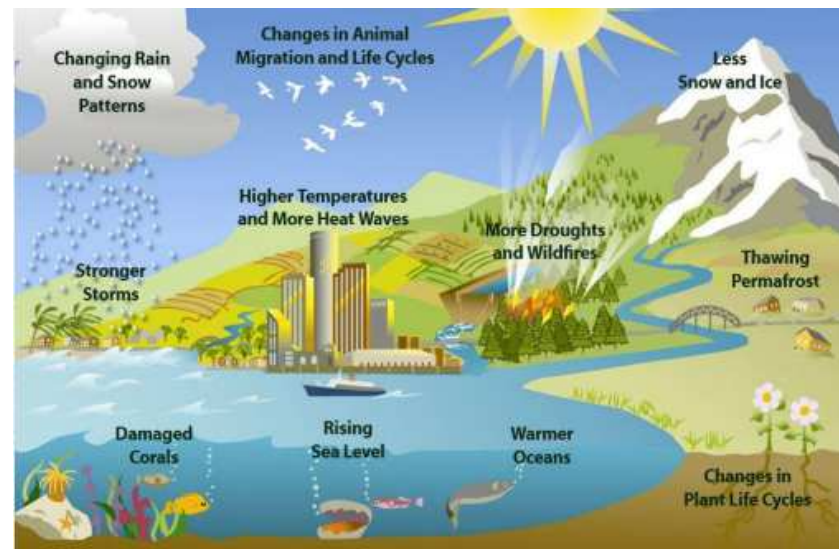
Sustainable Development Goals



Some of the major issues encompassed under environmental science include

➤ **Natural Resources – their conservation and management**

- Study the natural resources such as forests which contribute towards maintaining a balance in the environment. Managing and maintenance of forests and wildlife is an important task under natural resources conservation.



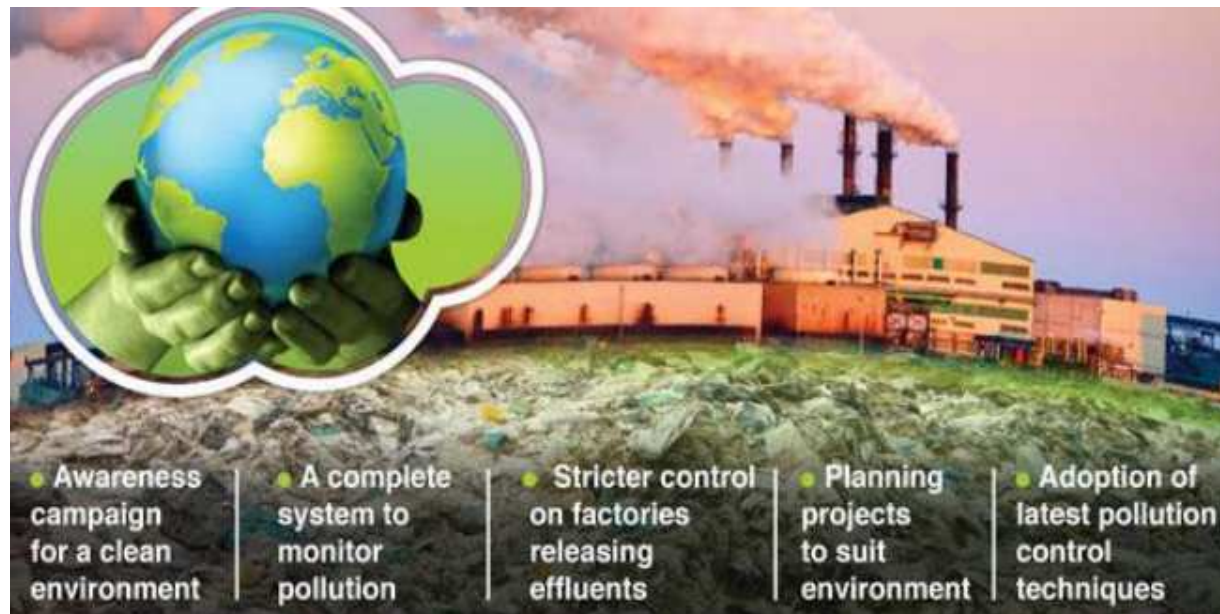
➤ Ecology and Biodiversity conservation



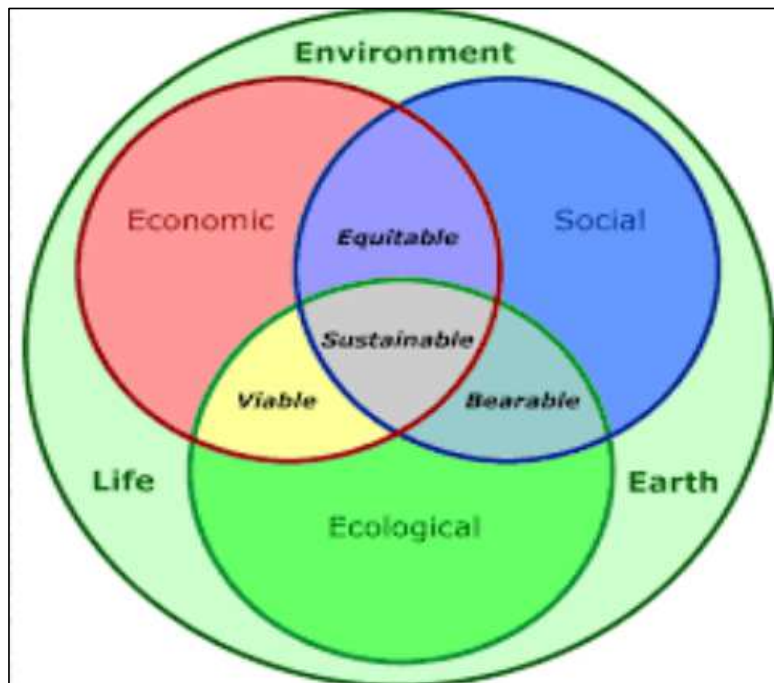
- Study the structure of ecosystem and its function.
- Study the importance of biodiversity in a healthy and sustainable ecosystem

➤ Environmental Pollution and Control

With the knowledge of environmental studies one can look for methods to control pollution and manage waste effectively



➤ Social issues in relation to development and environment



- **Study the holistic development**, i.e., improving global economy and standards of human life
- Simultaneously **study how to solve the social & environmental issues** like scarcity for clean water, demand for energy, consumerism, waste products, nuclear accidents, rehabilitation issues, acid rains, ozone depletion, climate change, global warming, population explosion etc.